

An Analytic Manipulation Bound for the Carry-Over Lottery Allocation Family

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Abstract

The National Basketball Association (NBA) draft lottery is a fair-allocation problem with a documented manipulation incentive: teams eliminated from playoff contention can improve their expected draft position by losing additional games. Munro and Banchio [2020] prove an impossibility result: no mechanism mapping end-of-season records to draft probabilities can simultaneously reward poor performance and eliminate this incentive. Carry-Over Lottery Allocation (Cola) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [Highley et al., 2026]. The mechanism’s incentive-compatibility argument relies on five assumptions, four of which hold by construction; the fifth—that pool-manipulation gains are negligible—is supported by simulation but lacks a closed-form bound. We make two contributions. First, we derive an analytic upper bound of approximately 4% on the worst-case probability gain from pool manipulation under empirically calibrated index distributions, with a closed-form sensitivity to the underlying parameters. Second, we present a brief historical backtest (26 NBA seasons, five variants) and an analytic per-series bound for Capped Cola, the variant introduced in response to recent NBA reform discussions. We also surface an open mapping between the Cola family and the axiomatic draft-rule literature [Coreno and Balbuzanov, 2022].

1 Introduction

The annual NBA draft allocates a small number of indivisible items (draft picks) to a fixed set of agents (the league’s 30 teams). Among the 30 teams, the 14 that miss the playoffs participate in a lottery for the top picks; the rest receive picks in reverse order of regular-season record. The lottery is the focus of our paper because it carries the policy weight: it determines who gets the franchise-altering top picks each year, and its design has

been revised three times in 60 years to address an empirically observed manipulation incentive known as tanking. A team eliminated from playoff contention can improve its expected draft position by losing additional games, and the design of any priority rule must contend with this.

Munro and Banchio [2020] formalise the tension as an impossibility. They define a No-Tanking Condition (NTC): no team can improve its expected outcome by losing. Under NTC and the requirement that worse-record teams receive strictly better draft odds, only a uniform lottery (equal odds for all non-playoff teams) survives—abandoning the goal of helping the weakest teams. The 2026 NBA reform proposal known as “3-2-1” takes precisely this route, with adjustments for play-in-tournament eligibility [Highley, 2026].

Carry-Over Lottery Allocation (Cola) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [Highley et al., 2026]. Each non-playoff team accumulates lottery tickets over consecutive seasons of missed playoffs; success in the playoffs or in the draft itself diminishes a team’s accumulated tickets. The single-season manipulation channel that drives the Munro-Banchio counterexample becomes ineffective because no single-season choice materially shifts the multi-year ticket pool.

Our contributions.

- An analytic manipulation bound. Cola’s incentive-compatibility argument rests on five assumptions, four of which hold by construction. The fifth—pool manipulation gains are negligible—is supported in the original paper by simulation over 1,000 synthetic seasons. We derive a closed-form upper bound of approximately 4% on the worst-case probability gain, with explicit sensitivity to the underlying index-distribution parameters (Section 4). The bound contains the simulation-observed extremes and provides the first parameter-explicit ceiling.
- A family-level perspective. We present the Cola mechanism family (Section 3) as five variants on a shared scaffold, including Capped Cola, the variant introduced in response to NBA-side feedback to

bound the per-series cost of advancing in the play-offs. We derive an analytic per-series bound for Capped Cola (Lemma 3). Empirical results from a 26-season historical backtest (Section 5) calibrate the bounds.

- A bridge to the axiomatic fair-division literature. Coreno and Balbuzanov [2022] establish an impossibility for draft rules satisfying Respect for Priority, Envy-Free up to One Object, and Strategy-Proofness simultaneously. We surface a preliminary mapping between Cola’s priority structure and these axioms (Section 2.2) as an invitation for further axiomatic analysis.

2 Background and Related Work

2.1 Effort Manipulation: The Munro-Banchio Impossibility

Munro and Banchio [2020] establish that any mechanism mapping end-of-season records to draft probabilities, while giving worse teams strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two teams, both eliminated, with identical records and a final game against each other. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon the goal of differential reward.

The authors propose an alternative, T-IC (Truncated Incentive Compatibility), which tracks each team’s conditional probability of finishing last and freezes allocations at the moment IC would first be violated. T-IC is optimal in theory but computationally expensive, opaque to fans, and limited to single-season scope; calculation requires precise team-level valuations of playoff-versus-draft trade-offs that the league does not have, and some teams have an incentive to tank from the first game of the season, which the truncation timing does not reach.

2.2 Preference Manipulation: The Coreno-Balbuzanov Impossibility

Coreno and Balbuzanov [2022] address a complementary manipulation vector: teams misreporting which players they prefer. They prove that no allocation rule simultaneously satisfies:

- Respect for Priority (RP): higher-priority teams receive weakly better allocations [Svensson, 1994];
- Envy-Free up to One Object (EF1): no team envies another’s allocation after removing one item [Lipton et al., 2004; Budish, 2011];
- Strategy-Proofness (SP): no team benefits from misreporting preferences.

The constituent axioms trace to peer-reviewed sources. RP extends Svensson’s “weak fairness” from queue allocation to draft priority [Svensson, 1994]. EF1 originates with [Lipton et al., 2004] and is adapted to the indivisible-goods assignment setting by [Budish, 2011].

Whether RP and EF1 lift from single-period draft rules to Cola’s multi-year mechanism is an open question. A preliminary reading: under Cola, longer droughts produce weakly better expected draft position, which mirrors RP if priority is identified with the multi-year ticket ranking; the graduated diminishment schedule preserves EF1 in the sense that no team prefers another’s outcome once one “lucky” lottery win is removed. We do not assert this lift; we surface it as an axiomatic question that the workshop’s audience is well placed to resolve.

3 The COLA Mechanism Family

3.1 Common Framework

Every Cola variant maintains a per-team, integer-valued state across seasons and uses that state to determine draft priority in the current season. We refer to this state as the lottery index and write $L_i^{(t)}$ for team i at the end of season t .

Definition 1 (Lottery Index). For each team i and season t , the lottery index $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$ is determined by an initial value $L_i^{(0)}$, a per-season increment $\Delta_i^{(t)}$ (variant-specific), and a per-season diminishment $\rho_i^{(t)} \in [0, 1]$ that scales the index down based on playoff and draft outcomes:

$$L_i^{(t+1)} = \left(L_i^{(t)} + \Delta_i^{(t)} \right) \cdot \left(1 - \rho_i^{(t)} \right).$$

The pool in season t is $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$, where $E^{(t)}$ is the set of teams eligible for the lottery.

The three components — the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$ — are the levers that distinguish the variants.

3.2 Five Variants

Simple Cola eliminates the lottery: eligibility is the 22 teams with zero playoff series wins (14 non-playoff teams + 8 first-round losers). Teams are ranked by drought (consecutive seasons without a playoff series win or top-3 lottery pick), tiebroken by most regular-season wins. Picks are assigned deterministically.

Simple Lottery Cola shares Simple’s drought state and 22-team pool. Pre-2019 NBA lottery odds (25.0% down to 0.5% across the worst 14 teams) are applied to the top 14 by drought; the remaining 8 receive picks 15–22 deterministically.

Classic Cola is the variant developed in detail by Hightley et al. [2026] and the variant our manipulation bound directly analyses. The eligibility set is the 14 non-playoff teams; each non-playoff team receives an increment of $\alpha = 1,000$ tickets per season; playoff teams receive zero increment. Diminishment is graduated: champion ($\rho = 1.0$), finals (0.75), conference finals (0.5), second round (0.25), first round (0); draft pick #1 ($\rho = 1.0$), #2 (0.75), #3 (0.5), #4 (0.25), #5 and below (0). The lottery for picks 1–4 is weighted by lottery index ($p_i = L_i/P$); picks 5–14 are assigned by lottery-index rank.

Countdown Cola ranks teams by McCarty number $M_i = d_i \cdot w_i$ (drought $\times$ regular-season wins). The same 22-team pool as Simple Cola is partitioned into nested 5-team elimination pools; draft picks are assigned bottom-up via ticket allocation (2, 3, 4, 5, 6) within each pool. The mechanism guarantees that no team falls more than four spots below its rank.

Capped Cola [Highley, 2026] caps the lottery index at Max (default 150). The increment is $\Delta_i^{(t)} = w_i^{(t)}$ for any team finishing the regular season outside the top 6 of either conference; eligibility excludes (i) teams that won a top-5 lottery pick last season and (ii) teams that won a playoff series this season or last. Diminishment uses a six-step playoff schedule (champion, runner-up, conference finals, second round, first round, play-in winner losing R1) and a five-step draft schedule, smoothing the cliffs of Classic Cola.

3.3 Incentive Compatibility

Highley et al. [2026] establish IC for Classic Cola under five assumptions:

1. Playoff primacy: the value to a team of advancing in the playoffs exceeds the value of any plausible draft-pick gain.
2. Playoff advancement: within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.
3. Picks 5–14 negligible at the margin: no team profits from manipulating its position among those slots through within-season effort.
4. Pool manipulation negligible: no team can meaningfully shift its expected draft position by manipulating which other teams are in or out of the lottery pool.
5. No traded-pick protections: pick trades do not interact with diminishment in a way that creates manipulation incentives.

Assumptions (1)–(3) and (5) hold by construction or follow from straightforward arguments. Assumption (4) is the open one: Highley et al. [2026] support it via 1,000-season simulation, observing a maximum probability gain of approximately 3%. The next section converts this empirical observation into a closed-form analytic bound.

4 Pool Manipulation Bound

4.1 Setup

Let $P = \sum_{i=1}^n L_i$ denote the total lottery pool, where L_i is team i 's lottery index and n is the number of lottery-eligible teams. A team's probability of winning the first pick is $p_i = L_i/P$.

The pool manipulation channel operates as follows. Team i deliberately loses to a high-index opponent h near the playoff boundary, causing h to make the playoffs. Team h (with index L_h) exits the lottery pool and is

replaced by a lower-index team ℓ (with index $L_\ell < L_h$). The pool shrinks by $\Delta = L_h - L_\ell > 0$.

Proposition 1 (Manipulation Gain). The probability gain for team i from a pool swap of size $\Delta = L_h - L_\ell$ is $G_i = \frac{L_i \cdot \Delta}{P \cdot (P - \Delta)}$.

Proof. Before manipulation, $p_i = L_i/P$. After the swap, the new pool is $P' = P - \Delta$ and $p'_i = L_i/P'$. The gain is $G_i = p'_i - p_i = L_i \left(\frac{1}{P - \Delta} - \frac{1}{P} \right) = \frac{L_i \cdot \Delta}{P \cdot (P - \Delta)}$. $\square$

When $\Delta \ll P$, the gain factors into two ratios: the manipulator's pool share (p_i) and the fractional pool change (Δ/P). Both have structural ceilings.

4.2 Conditioned Upper Bound

Definition 2 (Structural Anti-Correlation). Let $T_{\max}$ denote the maximum consecutive non-playoff years for any team, and T_{boundary} the maximum consecutive non-playoff years for a team near the playoff boundary (i.e., a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits structural anti-correlation when $T_{\text{boundary}} \ll T_{\max}$: teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.

Theorem 2 (Conditioned Manipulation Bound). Under structural anti-correlation with parameters $T_{\max}$, T_{boundary} , and average index multiplier $\bar{k} = \bar{L}/\alpha$, the maximum manipulation gain satisfies

$$G_{\max} \leq \frac{T_{\max} \cdot (T_{\text{boundary}} - 1)}{(T_{\max} + n - 1) \cdot n \cdot \bar{k}}.$$

Proof sketch. The gain is maximised when L_i is maximised and Δ/P is maximised. Bounding $p_{\max}$: the highest-index team has $L_{\max} = T_{\max} \cdot \alpha$. The minimum pool size occurs when one team has $T_{\max}$ years of accumulation and the remaining $n - 1$ teams each have the minimum α : $P_{\min} = (T_{\max} + n - 1) \cdot \alpha$. Thus $p_{\max} \leq T_{\max}/(T_{\max} + n - 1)$. Bounding Δ/P : the team being pushed out has at most $T_{\text{boundary}} \cdot \alpha$ tickets, and the team replacing it has at least α , so $\Delta_{\max} = (T_{\text{boundary}} - 1) \cdot \alpha$. In steady state, $P \approx n \cdot \bar{k} \cdot \alpha$. Combining via the first-order approximation yields the stated bound. $\square$

Empirical calibration. Using $T_{\max} = 10$ (the longest observed steady-state non-playoff stretch in our 26-season backtest), $T_{\text{boundary}} = 5$, $n = 14$, and $\bar{k} = 3.1$ (an average team has roughly three years of carry-over before reset, matching the original paper's observation), we obtain $G_{\max} \leq 10 \cdot 4/(23 \cdot 14 \cdot 3.1) \approx 4.0\%$. This contains the simulation-observed maximum of 3.0% across 1,000 seasons [Highley et al., 2026] with approximately 1.3 $\times$ headroom. An independent 200-season Bradley-Terry simulation we ran confirms the bound, with observed maximum gain of 1.70% (Figure 1).

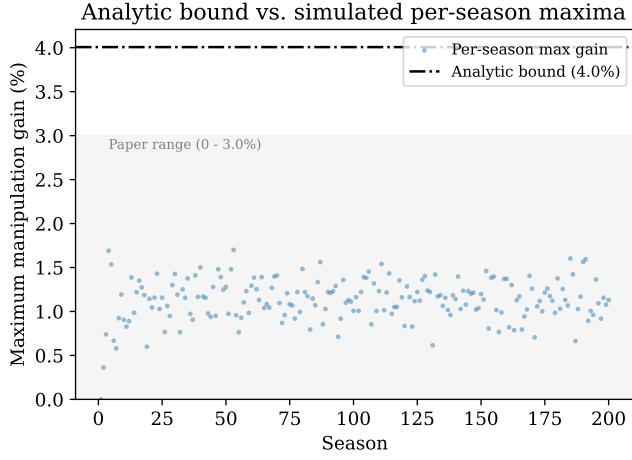


Figure 1: Analytic bound (red) vs. simulation distribution across 200 Bradley-Terry seasons. The bound contains all simulated outcomes with margin.

4.3 Per-Series Bound for Capped Cola

Capped Cola introduces a fourth manipulation channel: tanking through the playoffs themselves (declining to advance in order to retain stockpile). The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded above when the stockpile grows large) to a fraction-of-Max cost (bounded by construction).

Lemma 3 (Per-Series Tanking Cost). Let Max be the Capped Cola stockpile cap. For any team entering the playoffs with $L_i \leq \text{Max}$, the maximum reduction in L_i attributable to winning (rather than losing) any single playoff series is bounded by:

$$\Delta L_{\text{series}}^{\text{max}} = \begin{cases} 0.3 \cdot \text{Max} & \text{play-in advancers in round 1,} \\ 0.2 \cdot \text{Max} & \text{any other series advancement.} \end{cases}$$

At Max = 150, the bound is 45 tickets for the play-in case and 30 tickets for each subsequent series.

Proof. Reading off Capped Cola’s six-step diminishment schedule, the per-series increments in the diminishment fraction $\Delta\rho$ are: play-in advancer round 1 ($\rho_{\text{lose}} = 0.1$, $\rho_{\text{advance}} = 0.4$, $\Delta\rho = 0.3$); standard round 1 ($0.2 \rightarrow 0.4$, $\Delta\rho = 0.2$); round 2 ($0.4 \rightarrow 0.6$); conference finals ($0.6 \rightarrow 0.8$); finals ($0.8 \rightarrow 1.0$). All non-play-in transitions yield $\Delta\rho = 0.2$. The cost in tickets is $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \text{Max}$, with equality at the cap. $\square$

The bound makes the playoff-tanking incentive transparent at the league level: at Max = 150, no single series win ever costs a team more than 45 lottery tickets, which is approximately a year and a half of accumulation for a team near the wins-increment ceiling.

5 Empirical Validation

5.1 Backtesting Engine

We compute Cola state for every team across 26 NBA seasons (1999–2025), using verified Basketball Refer-

ence data: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilise by approximately season 5; we report aggregates over the 21 stable seasons (2005–2025).

5.2 Capped COLA Cap Calibration

Capped Cola is parameterised by a single tunable constant Max. The cap controls the trade-off between (i) ordering resolution at the top of the lottery (a higher cap permits more separation between the highest-stockpile teams) and (ii) the per-series tanking exposure of Lemma 3 (a higher cap raises the maximum cost of advancing past round 1). Table 1 reports the 21-stable-season aggregates from a sweep over $\text{Max} \in \{75, 100, 125, 150, 175, 200\}$.

Three patterns emerge. Sub-100 caps over-compress the lottery: at Max = 75, more than six teams sit at the cap on average and the rank-1 vs. rank-5 separation collapses to under 1 ticket. Caps above 175 erase the cap’s purpose: fewer than one team sits at the cap on average. The published Max = 150 choice produces fewer than two teams at the cap in 19 of 21 stable seasons and at most three in any single season, matching the variant’s published headline “no more than three teams at the maximum at a time, and on average fewer than two.”

5.3 Cross-Variant Comparison

The five variants rank teams differently under identical historical data. In 2024–25, the Sacramento Kings lead Classic Cola’s ticket pool (16-season drought, accumulated index 11,250) but Chicago leads Simple, Simple Lottery, and Countdown Cola (longest-running multi-year stretch by drought count alone). Capped Cola places multiple teams at the cap simultaneously, eliminating the differential between long-rebuilding franchises in a single year and trading resolution for a bounded per-series exposure.

6 Discussion

The bounds in this paper establish that Cola’s open assumption (4) is bounded by approximately 4% under empirically calibrated parameters, and that the additional manipulation channel introduced by Capped Cola’s cap is bounded at $0.3 \cdot \text{Max}$ tickets per series. Combined, the four manipulation channels available under Capped Cola—single-season tanking (zero, by construction), pool manipulation ($\sim 4\%$ probability gain), multi-year strategic tanking (bounded by the diminishment schedule), and per-series playoff tanking (45 tickets at Max = 150)—compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

The 2026 NBA reform proposal known as “3-2-1” adopts a different escape from the Munro-Banchio impossibility: it expands the lottery to 16 teams with a

Max	Teams at cap (mean)	Teams at cap (max)	Sep. R1–R5 (mean)	Per-series cost (max)	Cum. regret (max)
75	6.71	9	0.9	22.5	45.0
100	4.33	8	11.7	30.0	53.7
125	2.86	4	27.9	37.5	53.7
150	1.62	3	47.7	45.0	57.6
175	0.95	2	65.4	52.5	67.2
200	0.52	2	76.4	60.0	76.8

Table 1: Capped Cola aggregates across 21 stable seasons (2005–2025). Per-series cost max = $0.3 \cdot \text{Max}$ from Lemma 3; cumulative regret max is the worst empirical observation. The bolded Max = 150 row is the variant’s published default [Highley, 2026].

bounded-spread ball allocation that approximates the uniform lottery within an adjustment for the play-in tournament. The proposal includes a three-year re-evaluation provision [Highley, 2026]. Cola’s contribution to that re-evaluation conversation is the multi-year-history dimension that 3-2-1 omits, and the bounds in this paper formalise the manipulation properties that any future-iteration discussion will need.

Two open directions extend this work. First, the structural anti-correlation in Definition 2 is calibrated, not formally derived; a stationarity argument that bounds T_{boundary} as a function of league parameters would close Assumption (4) entirely. Second, the lift from Coreno and Balbuzanov [2022]’s axioms to multi-period priority mechanisms (Section 2.2) is open: whether Cola’s ticket ordering satisfies RP and EF1 in a formal sense is a fair-division question on which workshop participants are well placed to make progress.

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